6. Construct a C program to simulate Round Robin scheduling algorithm with C.

main.c

Run ▶

```c
#include <stdio.h>
int main() {
    int n, i, time = 0, tq;
    int bt[20], rt[20];
    printf("Enter number of processes: ");
    scanf("%d", &n);
    for (i = 0; i < n; i++) {
        printf("Enter burst time of P%d: ", i + 1);
        scanf("%d", &bt[i]);
        rt[i] = bt[i];
    }
    printf("Enter time quantum: ");
    scanf("%d", &tq);
    while (1) {
        int done = 1;
        for (i = 0; i < n; i++) {
            if (rt[i] > 0) {
                done = 0;
                if (rt[i] > tq) {
                    time += tq;
                    rt[i] -= tq;
                } else {
                    time += rt[i];
                    rt[i] = 0;
                    printf("P%d completed at time %d\n",
                           i + 1, time);
```

Output

```
Enter number of processes: 3
Enter burst time of P1: 5
Enter burst time of P2: 3
Enter burst time of P3: 1
Enter time quantum: 2
P3 completed at time 5
P2 completed at time 8
P1 completed at time 9
```

```c
                           i + 1, time);
                }
            }
        }
        if (done)
            break;
    }
    return 0;
}
```